Tutorial-6

Lasso Regression

$$sign(w_j) = \begin{cases} 1, & w > 0 \\ -1, & w < 0 \\ 0, & w = 0 \end{cases}$$

$$loss = error + \underbrace{\lambda |w|}_{penalty}$$

$$f(x) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$$

$$loss = \frac{1}{2n} \sum_{i=1}^{n} ((f(x) - y)^2) + \lambda |w|$$

$$\left\{ |w| \Rightarrow \sum_{i=1}^{n} |w_i| \right\}$$

weights updation:

$$w_0^{new} = w_0^{old} + \eta \left(\left(\frac{1}{n}\right) \sum_{i=1}^{n} (y_i - y_{ipred}) \right)$$

$$w_j^{new} = w_j^{old} + \eta \left(\left(\frac{1}{n}\right) \sum_{i=1}^{n} ((y_i - y_{ipred}) x_j) + \lambda \, sign(w_j) \right)$$

x_1	x_2	x_3	x_4	y_i
15	5.2	24	0	0.2
8	12.5	12	3	0.7
20	3.8	36	0	0.1
5	8.0	6	5	0.9
18	6.5	24	1	0.3

$w_0 = 0.5$, $w_1 = w_2 = w_3 = w_4 = 0$

$\eta = 0.01$, $\lambda = 0.5$

1st iteration

$\Rightarrow$ data $(15, 5.2, 24, 0, 0.2)$

$f_1(x) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$

$= 0.5 + 0 + 0 + 0 + 0 = 0.5$

$\Rightarrow$ data $(8, 12.5, 12, 3, 0.7)$

$f_2(x) = 0.5 + 0 = 0.5$

$\Rightarrow$ data $(20, 3.8, 36, 0, 0.1)$

$f_3(x) = 0.5 + 0 = 0.5$

$\Rightarrow$ data $(5, 8.0, 6, 5, 0.9)$

$f_4(x) = 0.5 + 0 = 0.5$

$\Rightarrow$ data $(18, 6.5, 24, 1, 0.3)$

$f_5(x) = 0.5 + 0 = 0.5$

$$\text{loss} = \frac{1}{2n}\left(\sum_{i=1}^{n}\left((f(x)-y)^2\right)\right) + \left(\lambda|w|\right)$$

$$= \frac{1}{2\times 5}\Big((0.5-0.2)^2 + (0.5-0.7)^2 + (0.5-0.1)^2 + (0.5-0.9)^2 + (0.5-0.3)^2\Big) + 0.5(0+0+0+0)$$

$$= \frac{1}{10}\left((0.3)^2 + (-0.2)^2 + (0.4)^2 + (-0.4)^2 + (0.2)^2\right)$$

$$= \frac{1}{10}\left(0.09 + 0.04 + 0.16 + 0.16 + 0.04\right)$$

$$= \frac{0.69}{10} = 0.069$$

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weights updation

$$w_{0new} = w_{0old} - \eta \frac{1}{n}\left(\sum_{i=1}^{n} (-y_i + \hat{y}_{pred})\right)$$

$$= 0.5 - \frac{0.01}{5}\Big((-0.2+0.5)+(-0.7+0.5)+(-0.1+0.5)$$
$$+(0.5-0.9)+(0.5-0.3)\Big)$$

$$= 0.5 - \frac{0.01}{5}\left(0.3 - \cancel{0.2} + \cancel{0.4-0.4} + \cancel{0.2}\right)$$

$$= 0.5 - \frac{0.003}{5} = \cancel{0.5006}\ 0.4994$$

$$w_{1new} = w_{1old} - \eta\left(\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_{pred} - y_i)x_i + \lambda\, sign(w_1)\right)$$

$$= 0 - 0.01\Big(\frac{1}{5}\Big((0.5-0.2)15 + \cancel{(0.5-0.7)}8$$
$$+\cancel{(0.5-0.1)}20 + \cancel{(0.5-0.9)}5$$
$$+\cancel{(0.5-0.3)}18) + \cancel{0.5(0)}\Big)$$

$$= -0.01\Big(\frac{1}{5}\Big(0.3\times15 - 0.2\times8 + 0.4\times20$$
$$-0.4\times5 + 0.2\times18\Big)$$

$$= -\frac{0.01}{5}\Big(\overset{4.5}{\cancel{0.3\times15}} + \overset{2}{\cancel{0.2\times10}} + \overset{6}{\cancel{0.4\times15}}\Big)$$

$$= -0.025$$

$$w_{2new} = 0 - 0.01\Big(\frac{1}{5}\Big(0.3\times5.2 - 0.2\times12.5$$
$$+0.4\times3.8 - 0.4\times8$$
$$+0.2\times6.5\Big) + 0.5\times0\Big)$$

$$= -\frac{0.01}{5}\left(0.3\times5.2 - 0.2\times6 - 0.4\times4.2\right)$$

$$= -\frac{0.01}{5}(-1.32) = 0.00264$$

$$w_3^{new} = 0 - 0.01\left(\frac{1}{5}\left(0.3\times 24 - 0.2\times 12 + 0.4\times 36 - 0.4\times 6 + 0.2\times 24\right) + 0.5(0)\right)$$

$$= -0.0432$$

$$w_4^{new} = 0 - 0.01\left(\frac{1}{5}\left(\cancel{0.3\times 0} - 0.2\times 3 + 0.4\times 0 - 0.4\times 5 + 0.2\times 1\right) + 0.5(0)\right)$$

$$= -\frac{0.01}{5}\left(\overset{-0.4}{\cancel{-0.6}} - 2 + \cancel{0.2}\right) = \frac{0.01\times 2.4}{5}$$

$$= 0.0048$$

$w_0 = 0.4994$, $w_1 = -0.025$, $w_2 = 0.00264$, $w_3 = -0.0432$

$w_4 = 0.0048$

2nd iteration

⇒ data (15, 5.2, 24, 0, 0.2)

$$f_1(x) = 0.4994 - 0.025\times 15 + 0.00264\times 5.2 - 0.0432\times 24 + 0.0048\times 0$$

$$= \cancel{0.4994} - 0.898672$$

⇒ data (8, 12.5, 12, 3, 0.7)

$$f_2(x) = 0.4994 - 0.025\times \overset{8}{\cancel{15}} + 0.00264\times 12.5 - 0.0432\times 12 + 0.0048\times 3$$

$$= -0.1715$$

⇒ data (20, 3.8, 36, 0, 0.1)

$$f_3(x) = 0.4994 - 0.025\times 20 + 0.00264\times 3.8 - 0.0432\times 36 + 0.0048\times 0$$

$$= -1.545768$$

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$\Rightarrow$ data (5, 8.0, 6.5, 0.9)

$$f_4(x) = 0.4994 - 0.025 \times 5 + 0.00264 \times 8 - 0.0432 \times 6 + 0.0048 \times 5$$

$$= 0.16032$$

$\Rightarrow$ data (18, 6.5, 24, 1, 0.3)

$$f_5(x) = 0.4994 - 0.025 \times 18 + 0.00264 \times 6.5 - 0.0432 \times 24 + 0.0048 \times 5$$

$$= -0.96544$$

$$\text{loss} = \frac{1}{2 \times 5}\Big((-0.898672 - 0.2)^2 + (-0.1715 - 0.7)^2 + (-1.545768 - 0.1)^2 + (0.16032 - 0.9)^2 + (-0.96544 - 0.3)^2 \Big) \cancel{+ 0.5(}$$

$$+ 0.5\,(0.025 + 0.00264 + 0.0432 + 0.0048)$$

$$= \frac{1}{10}\Big((1.098672)^2 + (0.8715)^2 + (1.645768)^2 + (0.73968)^2 + (1.26544)^2 \Big)$$

$$+ 0.5\,(0.07564)$$

$$= \frac{1}{10}(6.8236) + 0.5\,(0.07564)$$

$$= 0.68236 + 0.03782$$

$$\text{loss} = 0.72018$$